

△ Show that $\sigma'(y) = \sigma(y)(1 - \sigma(y)) \Rightarrow$

we know that $\sigma(y) = \frac{1}{1+e^{-y}}$ in sigmoid activation

function $\Rightarrow \boxed{y = w \cdot x}$

$$\sigma'(y) = \frac{d}{dy} \sigma(y) = \frac{d}{dy} \left(\frac{1}{1+e^{-y}} \right) \quad \text{based on derivation}$$

$$\sigma'(y) = \frac{(1+e^{-y}) \cdot 0 - 1(-e^{-y})}{(1+e^{-y})^2} = \frac{e^{-y}}{(1+e^{-y})^2}$$

now we have to simplify this $\Rightarrow$

$$= \frac{1 \cdot e^{-y}}{(1+e^{-y})(1+e^{-y})} = \frac{1}{(1+e^{-y})} \cdot \frac{e^{-y}}{(1+e^{-y})}$$

$$= \frac{1}{(1+e^{-y})} \cdot \frac{e^{-y} + 1 - 1}{(1+e^{-y})} = \frac{1}{\underbrace{1+e^{-y}}_{\sigma(y)}} \cdot \left(1 - \frac{1}{\underbrace{1+e^{-y}}_{\sigma(y)}} \right)$$

$$= \sigma(y)(1 - \sigma(y))$$

$$\Rightarrow \sigma'(y) = \sigma(y)(1 - \sigma(y))$$

we prove! 😊